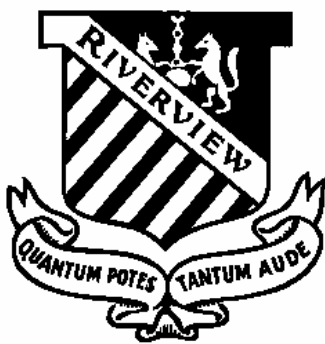




SAINT IGNATIUS' COLLEGE

Trial Higher School Certificate

2010

MATHEMATICS

Directions to Students

• Reading Time : 5 minutes	• Total Marks 120
• Working Time : 3 hours	
• Write using blue or black pen. (sketches in pencil).	• Attempt Question 1 – 10
• Board approved calculators may be used	• All questions are of equal value
• A table of standard integrals is provided at the back of this paper.	
• All necessary working should be shown in every question.	
• Answer each question in the booklets provided and clearly label your name and teacher's name.	

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Question 1 (Start a new Booklet)

Marks

- (a) Calculate $e^{3.1}$ correct to 2 significant figures 2
- (b) Solve for x : $x^2 + 5x = 24$ 2
- (c) Find the primitive of $(2x + 5)^4$ 2
- (d) Calculate the exact value of $\cos \frac{\pi}{6}$ 2
- (e) Expand and simplify $(3 + 2\sqrt{2})^2$ 2
- (f) Find the sum of the first 40 terms of the series $4 + 10 + 16 + \dots$ 2

Question 2 (Start a new Booklet)

Marks

- (a) Find the derivative of
- (i) $x \sin 3x$ 2
- (ii) $\frac{e^x}{x}$ 2
- (b) Integrate the following
- (i) $\frac{3x}{x^2 + 4}$ 2
- (ii) $3 \sec^2 \frac{x}{2}$ 1
- (c) Find the equation of the tangent to the curve $y = x^3 + 1$ at the point where $x = 1$. 3
- (d) Evaluate $\sum_{n=1}^3 n^3 + 3$ 2

Question 3 (Start a new Booklet)**Marks**

- (a) $A(1, -4)$ is a point on the line $J: 3x + 2y + 5 = 0$
- (i) Show that the point $B(-3,2)$ lies on the line. 1
 - (ii) Find the equation of the line perpendicular to J passing through the point $C(3,1)$. 2
 - (iii) Calculate the distance AB . 2
 - (iv) Find the perpendicular distance from C to the line J . 2
 - (v) Calculate the area of $\triangle ABC$. 1
- (b) The Pacific Star cruise ship travels 215 km on a bearing of 085° from Sydney. It then travels 112km on a bearing of 135° .
- (i) How far is the ship from Sydney? 2
 - (ii) What is the final bearing of the ship from Sydney?
(give answer correct to the nearest degree) 2

Question 4 (Start a new Booklet)**Marks**

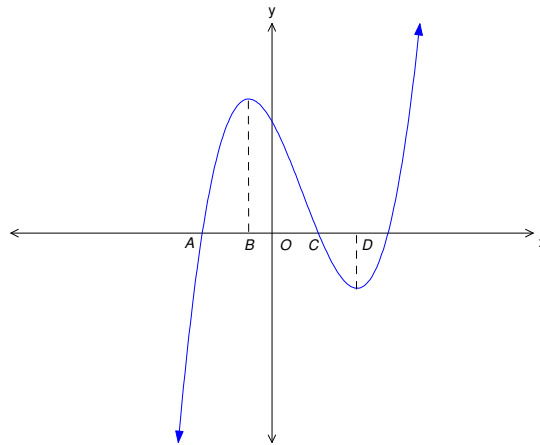
- (a) For the parabola $x^2 + 4x - 12y + 40 = 0$:
- (i) Use completing the square method to write the equation in the form $(x - h)^2 = 4a(y - k)$ 1
 - (ii) Find the focal length. 1
 - (iii) Write down the coordinates of the vertex. 1
 - (iii) Find the focus. 1
- (b) Solve $\log x + \log (x + 4) = \log 12$ 2
- (c) William drops a ball out of a window that is 25 m above the ground. On the first rebound, it rises to a height of 20 m. On subsequent rebounds, it rises to a height equal to $\frac{4}{5}$ of its previous height. If there is no interference with the ball, calculate the total distance through which the ball moves before coming to rest. 2
- (d) At a Primary School Sports Carnival the combined year relay race has one participant from each year group from Kinder to Year 6. The Kinder child runs 15 m to a point and returns to the start, then the year 1 student runs 20 m and returns to the start. Each child runs in turn with each year group running 5 m further than the previous year group.
- (i) How far does the year 6 child have to run? 2
 - (ii) How far is run by the students in one complete race? 2

Question 5 (Start a new Booklet)

Marks

- (a) Copy the following diagram into your answer booklet.

2



The curve represents the function $y = f(x)$. On the same set of axes draw the derivative function $y = f'(x)$.

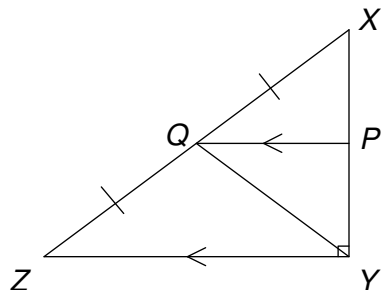
- (b) Consider the function $y = 2x^3 - 9x^2 + 12x$.

- | | | |
|-------|--|---|
| (i) | Show that the only x -intercept exists at the origin. | 1 |
| (ii) | Find the stationary points and determine their nature. | 3 |
| (iii) | Show that a point of inflexion exists at the point $(1.5, 4.5)$. | 1 |
| (iv) | For what values of x is the curve monotonically decreasing? | 1 |
| (v) | Sketch the curve of the function $y = 2x^3 - 9x^2 + 12x$ in the domain $0 \leq x \leq 3$. | 2 |
| (vi) | Find the values of k for which $2x^3 - 9x^2 + 12x = k$ has only one solution. | 2 |

Question 6 (Start a new Booklet)

Marks

(a)



In the diagram above $\triangle XYZ$ is right angled. PQ is parallel to YZ and Q is the midpoint of XZ .

- | | | |
|-------|---|---|
| (i) | Copy the diagram into your answer booklet. | |
| (ii) | Give a reason why $\angle XPQ = 90^\circ$. | 1 |
| (iii) | Prove that $\triangle XPQ \equiv \triangle YPQ$. | 2 |
| (iv) | Prove $QZ = QY$. | 1 |
| | | |
| (b) | A cylindrical tank is filled with water. The volume of water in the tank is determined by the function $V = 7t^3 + 15t^2 - 3t$, where t is time in seconds and the volume in litres. What is the rate of change of the volume of the tank after 12 seconds have elapsed? | 2 |
| | | |
| (c) | Find the exact length of the radius of a circle in which an arc length of 10 cm subtends an angle of 50° at the centre of the circle. | 2 |
| | | |
| (d) | Given that α and β are the roots of the quadratic equation $2x^2 - 7x + 5 = 0$, find ; | |
| (i) | $\alpha + \beta$ | 1 |
| (ii) | $\alpha\beta$ | 1 |
| (iii) | $\alpha^2 + \beta^2$ | 2 |

Question 7 (Start a new Booklet)

Marks

(a) Prove that $\frac{\sin \theta}{1 + \cos \theta} = \frac{1 - \cos \theta}{\sin \theta}$

3

(b) The area bound by the curve $y = \sqrt{\sin x}$, $x = 0$, $x = \frac{\pi}{3}$ and the x -axis is rotated about the x -axis. Find the volume of the solid formed.

3

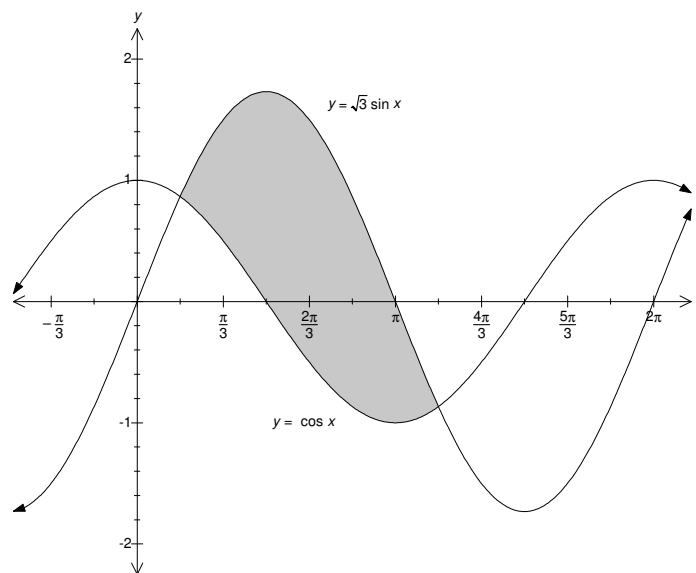
(c) A function $f(x)$ has a table of values

2

x	0	0.5	1	1.5	2
f(x)	3.24	4.16	2.25	1.15	0

Use Simpson's Rule to calculate $\int_0^2 f(x) dx$ (correct to 2 decimal places)

(d)



The graph above shows the sketch of the curves $y = \sqrt{3} \sin x$ and $y = \cos x$.

(i) Solve the curves simultaneously to show that the x -values of the points of intersection are $\frac{\pi}{6}$ and $\frac{7\pi}{6}$.

1

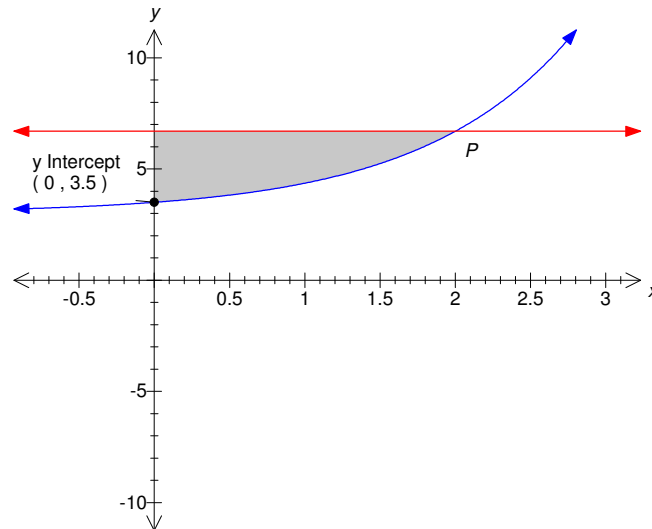
(ii) Find the shaded area in the diagram.

3

Question 8 (Start a new Booklet)

Marks

(a)



- | | | |
|-------|--|---|
| (i) | The curve $f(x)$ displayed above is in the form $f(x) = ae^x + 3$.
Given that the curve passes through the point $(0, 3.5)$, show that
$a = 0.5$. | 1 |
| (ii) | The x -value of the point P on the curve is 2. What is the y -value of
the point P ? | 1 |
| (iii) | Find the area of the shaded region. | 2 |
- (b) A particle moves such that its position, x metres, from a fixed point O is
given by the function $x = t^3 - 7\frac{1}{2}t^2 + 18t + 2$, where t is measured in
seconds.
- | | | |
|-------|---|---|
| (i) | Find the particle's initial position and velocity. | 3 |
| (ii) | When is the particle at rest? | 1 |
| (iii) | What is the acceleration of the particle when it is first at rest? | 2 |
| (iv) | Find the distance travelled by the particle in the first three seconds. | 2 |

Question 9 (Start a new Booklet)**Marks**

- (a) Jenni invests \$30000 into an account on the 1st of March. She receives 9% p.a. interest compounded monthly. On the first day of each month after that she withdraws \$250 immediately after the interest is paid.
- (i) How much money did she have in the account immediately after making the first withdrawal? 1
- (ii) Show that after making the n th withdrawal the balance of the account is given by $\left(33\,333\frac{1}{3} - 3\,333\frac{1}{3} \times 1.0075^n\right)$ 2
- (iii) Find the number of withdrawals that Jenni can make before there is no money left in the account. 2
- (b) An ant colony has a population that is described by the function $P = P_0 e^{kt}$. If the ant colony initially had 300 ants and after 100 days the population had increased to 550 ants, find:
- (i) the value of P_0 and k . 3
- (ii) the time taken for the population of the colony to reach 1000 ants (write your answer correct to the nearest day). 2
- (c) It was found that on the 1st of June, 30 students in a school had the flu. Over the next month the number of cases of students being sick with the flu increased at decreasing rate. Draw a graph that would describe this situation. 2

Question 10 (Start a new Booklet)**Marks**

- (a) Find the derivative of the function $y = x \log_e x$ and hence find $\int \log_e x \, dx$. 2
- (b) A sector of a circle has an area of $\frac{3\pi}{4} \text{ cm}^2$, while its arc length is $\frac{\pi}{4} \text{ cm}$. Find the radius and angle of the sector. 4
- (c) A cylindrical can is to hold $20\pi \text{ m}^3$. The material for the top and bottom costs $\$10/\text{m}^2$ and material for the side costs $\$8/\text{m}^2$.
- (i) Show that the total cost of the material for the can be expressed by the formula $C = 20\pi r^2 + 16\pi r h$ 1
- (ii) Show that $h = \frac{20}{r^2}$ 1
- (iii) Find an expression for the cost in terms of r and hence find the values of r and h such that the cost of the materials is a minimum. 4

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

2010 Riverview Mathematics Trials – Solutions

Question 1:

(a) 22 (2 significant figures)

$$\begin{aligned} \text{(b)} \quad x^2 + 5x - 24 &= 0 \\ (x + 8)(x - 3) &= 0 \\ x &= -8 \text{ or } x = 3 \end{aligned}$$

$$\text{(c)} \quad y = \frac{(2x+5)^3}{10} + C$$

$$\text{(d)} \quad \cos \frac{\pi}{6} = \frac{\sqrt{3}}{2}$$

$$\text{(e)} \quad 9 + 12\sqrt{2} + 8 = 17 + 12\sqrt{2}$$

$$\begin{aligned} \text{(f)} \quad S_n &= \frac{n}{2}(2a + 6(n-1)) \\ \therefore S_{40} &= 20(8 + 6 \times 39) = 4840 \end{aligned}$$

Question 2:

(a) (i)

$$\text{Let } u = x \text{ therefore } \frac{du}{dx} = 1$$

$$\text{Let } v = \sin 3x \text{ therefore } \frac{dv}{dx} = 3 \cos 3x$$

$$\text{Since } \frac{dy}{dx} = u'v + v'u \text{ we have:}$$

$$\frac{dy}{dx} = \sin 3x + 3x \cos 3x$$

(ii)

$$\text{Let } u = e^x \text{ therefore } \frac{du}{dx} = e^x$$

$$\text{Let } v = \frac{1}{x} \text{ therefore } \frac{dv}{dx} = -\frac{1}{x^2}$$

$$\text{Since } \frac{dy}{dx} = u'v + v'u \text{ we have:}$$

$$\begin{aligned} \frac{dy}{dx} &= \frac{e^x}{x} - \frac{e^x}{x^2} \\ &= \frac{e^x}{x^2}(x-1) \end{aligned}$$

(b) (i)

$$\begin{aligned} &\int \frac{3x}{x^2 + 4} dx \\ &= \frac{3}{2} \ln |x^2 + 4| + C \end{aligned}$$

(ii)

$$\begin{aligned} &\int 3 \sec^2 \frac{x}{2} dx \\ &= 6 \tan \frac{x}{2} + C \end{aligned}$$

$$\text{(c)} \quad \frac{dy}{dx} = 3x^2$$

$$\text{Therefore at } x = 1, m_T = 3$$

$$y - 2 = 3(x - 1)$$

$$y = 3x - 1 \text{ (or in general form: } 3x - y - 1 = 0)$$

$$\text{(d)} \quad (1 + 3) + (8 + 3) + (27 + 3) = 4 + 12 + 30$$

Q 3 a)

A(1, -4) $3x + 2y + 5 = 0$ ←

i) sub $x = -3, y = 2$ into J:

LHS:

$3(-3) + 2(2) + 5 \Rightarrow$

$-9 + 4 + 5 = 0 = RHS$

∴ ~~P~~ P(-3, 2) lies on the line.

Let line be R

ii) $\frac{m_R}{m_J} = -1$

∴ $m_J = -\frac{3}{2}$
C(3, 1)

$m_R = \frac{2}{3}$

$y - 1 = \frac{2}{3}(x - 3)$

$3y - 3 = 2x - 6$

∴ $2x - 3y - 3 = 0$

iii) C(3, 1)

~~a~~ a = 3 x = 3

b = 2 y = 1

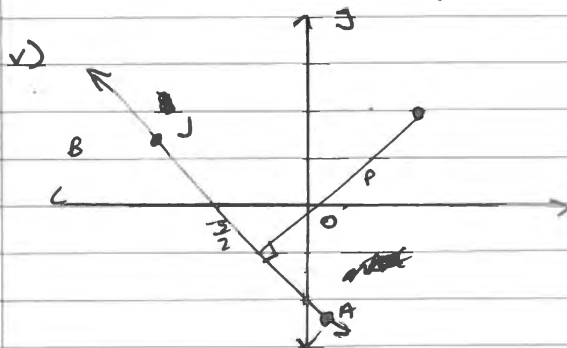
c = 5

$P = \frac{|a^2 + b^2 + c|}{\sqrt{a^2 + b^2}} = \frac{|3(3) + (2)(1) + 5|}{\sqrt{3^2 + 2^2}}$

$= \frac{9 + 2 + 5}{\sqrt{13}}$

$P = \frac{16}{\sqrt{13}}$ units

$P = \frac{16}{\sqrt{13}}$



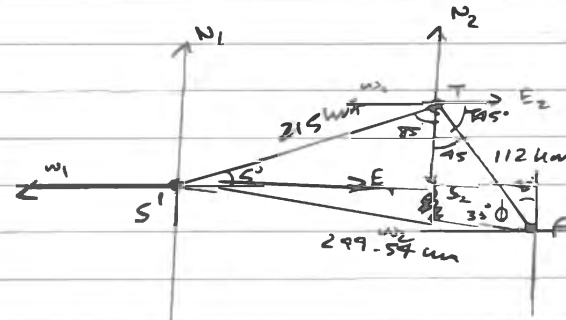
$d_{AB} = 2\sqrt{13}$

$d_P = \frac{16}{\sqrt{13}}$

$A = \frac{1}{2}ab$

$= \frac{1}{2} \cdot \frac{16}{\sqrt{13}} \cdot 2\sqrt{13} = 16$

b)



Let $S' = \text{Sydney}$

T = Turn

F = Stop

Join $S'F$

$\angle FTS = 45^\circ$ (adjacent angles add up to 180°)

$\angle S'TS = 85^\circ$ (adjacent angles of parallel lines are eq. & c.)

$\angle S'TF = 85 + 45 = 130^\circ$

$S'F = ?$

$a^2 = b^2 + c^2 - 2bc \cos \phi$

$(S'F)^2 = (S'T)^2 + (TF)^2 - 2(TF)(S'T) \cos \phi$

$(S'F)^2 = (215)^2 + (112)^2 - 2(112)(215) \cos 130^\circ$

$= 58769 - 9816 \cos 130^\circ$

$S'F = \sqrt{89725.65} = 299.5424031 \text{ km}$

$S'F = 299.54 \text{ km (2 d.p.)}$

ii) Let $\angle TFS' = \phi$

$\frac{\sin \phi}{215} = \frac{\sin 130}{299.54}$

$\sin \phi = \frac{215 \sin 130}{299.54}$

$\sin \phi = \frac{215 \sin 130}{299.54}$

$\phi = 33^\circ 21'$

$\angle S'FW_2 = 45^\circ - 33^\circ 21'$

$= 11^\circ 39'$

Bearing is

$270^\circ + 11^\circ 39'$

$= 282^\circ \text{ (measured from)}$

Q 4c)

$$x^2 + 4x - (12y + 40) = 0$$

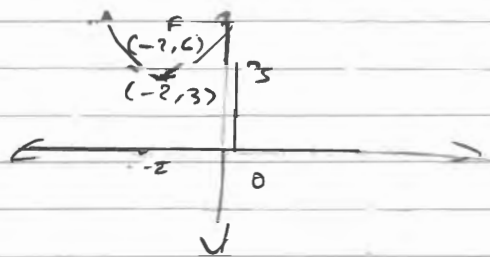
$$(x+2)^2 - 4 - 12y + 40 = 0$$

$$(x+2)^2 - 12y = -36$$

$$(x+2)^2 = -36 + 12y$$

$$(x+2)^2 = 12(y-3)$$

$$(x+2)^2 = 12(y-3)$$



$$4a = 12$$

$$|a| = 3$$

i) $4a = 12$
 $a = 3$

ii) $(-2, 3)$

iv) $(-2, 6)$

b) $\ln|x| + \ln|x+4| = \ln(12)$

$$\ln|x(x+4)| = \ln(12)$$

$$x^2 + 4x - 12 = 0$$

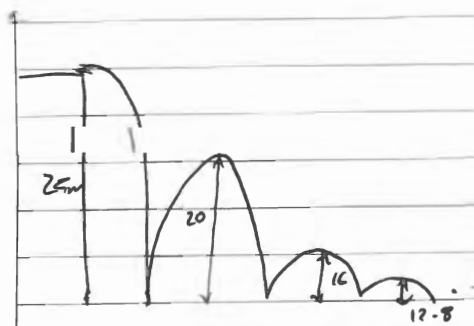
$$(x+6)(x-2) = 0$$

$$x = 2, x = -6$$

But $\ln|x|$, $x > 0$

$$\therefore x \neq -6$$

$$\{x = 2\}$$



$$T_n = \{40, 32, 25.6, \dots\} \rightarrow \{40, 32, 25.6, \dots\}$$

Let us define as $T_n \rightarrow 0$, it's at rest

$$T_n \text{ is a GP, } r = \frac{T_2}{T_1} = \frac{T_3}{T_2} = 0.8$$

$$T_n = ar^{n-1}$$

$$T = 40(0.8)^{n-1}$$

But window is 25m let define sum R

$$R = 25 + T_n$$

$$\text{Since } |r| < 0.8$$

S_{∞} exist

$$S_{\infty} = \frac{a}{1-r}$$

$$= \frac{40}{1-0.8} = 200 \text{ m}$$

$$R_n = 200 + 25 = 225 \text{ m}$$

$\therefore$ The ball travelled 225m



$$i) T_n = \{30, 40, 50, \dots, 90\}$$

$$T_n = a + (n-1)d$$

$$\text{Let } a = 30$$

$$d = 10$$

$$T_n = 30 + (n-1)10$$

$$= 30 - 10 + 10n$$

$$= 20 + 10n$$

$$\text{Let } n = 7$$

$$T_7 = 20 + 10(7) = 90m$$

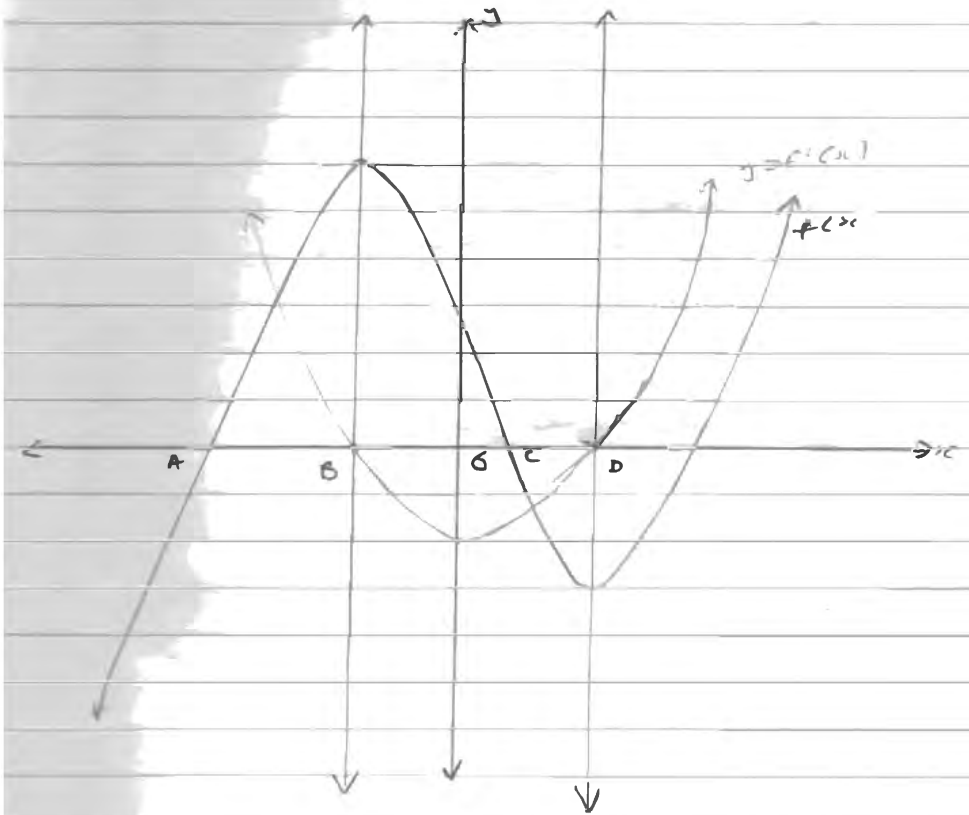
$$ii) S_n = T_1 + T_2 + \dots + T_n$$

$$S_n = \frac{n(a+l)}{2}$$

$$= \frac{7}{2} (30 + 90) = 420m$$

$\therefore$ 7 students run 420m altogether

Question 5 (a)



$$b) i) y = 2x^3 - 9x^2 + 12x$$

$$y' = 6x^2 - 18x + 12$$

$$\text{Let } y' = 0 \text{ (TP)}$$

$$0 = x^2 - 3x + 2$$

$$0 = (x-2)(x-1)$$

$$x = 2, x = 1$$

$$y = 4, y = 5$$

$$i) y = x(2x^2 - 9x + 12)$$

$$\text{Let } y = 0, x = 0, 2x^2 - 9x + 12 = 0$$

$$\text{But } \Delta = -15 (< 0)$$

$\therefore$ no real roots

continued
ii) ~~1.5~~

$$y'' = 12x - 18$$

$$\text{At } x = 1 \text{ in } y'$$

$$y'' = 12 - 18 = -6 (< 0) \text{ Maxime}$$

$$P_1(1, 5) \text{ TP (Max)}$$

$$\text{At } x = 2 \text{ in } y''$$

$$y'' = 12(2) - 18 = 6 (> 0) \text{ Minime}$$

$$P_2(2, 4) \text{ (TP) (Min)}$$

$$\therefore P_1(1, 5) \text{ (TP) (Max)}$$

$$P_2(2, 4) \text{ (TP) (Min)}$$

iii) At $y' = 0$ (IP)

$$0 = 12x - 18$$

$$x = \frac{18}{12} = 1.5$$

$$\text{At } y = 4.5$$

$$IP(1.5, 4.5)$$

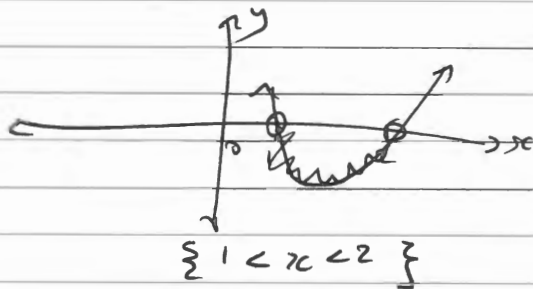
iv)

$$\frac{dy}{dx} < 0$$

$$y' = 6x^2 - 18x + 12$$

$$6x^2 - 18x + 12 < 0$$

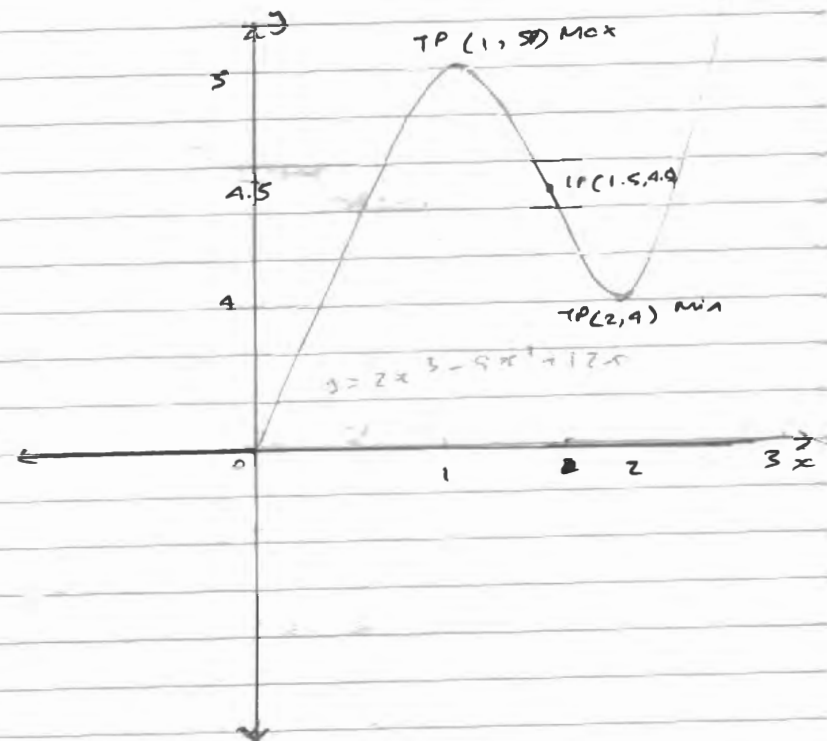
$$2 \cdot (x-2)(x-1) < 0$$



$\therefore$ graph is decreasing at $1 < x < 2$

v)

(3.9)



$$vi) 2x^3 - 9x^2 + 12x - k = 0$$

~~RMV~~
 $k = 0$ is a possible soln

$$\{ k < 4, k > 5 \}$$

Question 6

c)



ii) $\angle XPQ = 90^\circ = \angle PYZ$ (corresponding angles of parallel lines are equal)

iii) In $\triangle QXP$ and $\triangle XZY$:

$\angle QXP$ is common angle (INCLUDED)

$\angle XPQ = \angle XYZ = 90^\circ$ (angle)

$\therefore \triangle QXP \sim \triangle XZY$ (AAA)

$$\frac{XQ}{QZ} = \frac{XP}{PY} = 1 \text{ (common ratio of similar triangles)}$$

$$\therefore XP = PY$$

In $\triangle XQP$ and $\triangle YPQ$,

QP is common side (SIDE)

$XP = YP$ (SIDE)

$\angle QPY = \angle QPX = 90^\circ$ (INCLUDED ANGLE)

$\therefore \triangle XQP \cong \triangle YPQ$ (SAS)

iv) $QX = QY = QZ$ (given, equal side of congruent triangles)

$$\therefore QZ = QY$$

$$b) V = 7t^3 + 15t^2 - 3t$$

$$\frac{dV}{dt} = 21t^2 + 30t - 3$$

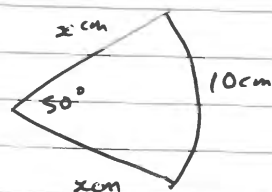
$$\text{At } t = 12$$

$$\frac{dV}{dt} = 21(12)^2 + 30(12) - 3$$

$$= 3381 \text{ L/s}$$

$$\therefore \frac{dV}{dt} = 3381 \text{ L/s at } t = 12 \text{ seconds}$$

c)



$$L = r\theta$$

$$10 = x \left(\frac{50\pi}{180} \right)$$

$$10 = x \left(\frac{5\pi}{18} \right)$$

$$\frac{180}{5\pi} = x$$

$$x = \frac{36}{\pi} \text{ cm}$$

$$d) i) A+B = \frac{-b}{a} = \frac{7}{2}$$

$$ii) AB = \frac{c}{a}$$

$$iii) A^2 + B^2 = (A+B)^2 - 2AB$$

$$= \left(\frac{7}{2}\right)^2 - 2\left(\frac{7}{2}\right)$$

$$= \frac{29}{4}$$

Question 7

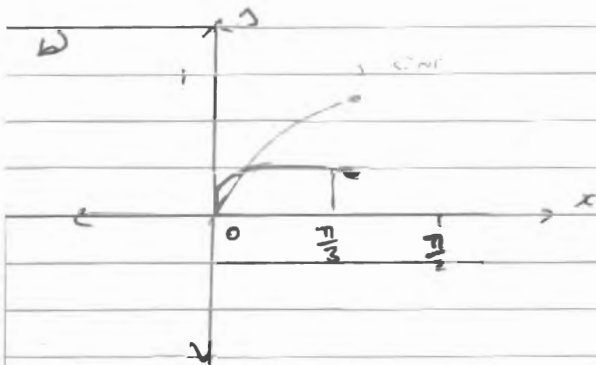
$$c) \frac{\sin \theta}{1 + \cos \theta} = \frac{1 - \cos \theta}{\sin \theta}$$

LHS:

$$\frac{\sin \theta (1 - \cos \theta)}{(1 + \cos \theta)(1 - \cos \theta)}$$

$$= \frac{\sin \theta (1 - \cos \theta)}{1 - \cos^2 \theta} \quad 1 - \cos^2 \theta = \sin^2 \theta$$

$$= \frac{\sin \theta (1 - \cos \theta)}{\sin^2 \theta} = \frac{1 - \cos \theta}{\sin \theta} = \text{RHS QED}$$



$$V_y = \pi \int_0^{\pi/3} y^2 dx$$

$$= \pi \int_0^{\pi/3} \sin^2 x dx$$

$$= \pi \left[\cos x \right]_0^{\pi/3}$$

$$= \pi \left[\cos \frac{\pi}{3} - \cos 0 \right]$$

$$= \pi \left[\frac{1}{2} - 1 \right]$$

$$= -\frac{\pi}{2}$$

$$2) \int_0^1 f(x) dx$$

$$A = \frac{1}{3} (f(1) + f(0) + 2 \text{ odd} + 4 \text{ even})$$

$$= \frac{0.5}{3} (0 + 3.24 + 2(2.25) + 4(4.16 + 1.15))$$

$$= \frac{0.5}{3} (28.98) = 4.83 \text{ (2dp)}$$

$$d) i) y = \sqrt{3} \sin x \quad (1)$$

$$y = \cos x \quad (2)$$

$$\frac{(1)}{(2)}$$

$$1 = \sqrt{3} \sin x$$

$$\tan x = \frac{1}{\sqrt{3}} \quad \frac{S(1)}{C}$$

$$x = \frac{\pi}{6}$$

So:

$$x = \frac{\pi}{6}$$

3rd:

$$x = \frac{\pi}{6} + \frac{\pi}{6}$$

$$x = \frac{2\pi}{6}$$

$$\therefore x = \frac{\pi}{6}, \frac{2\pi}{6} \text{ QED}$$

$$ii) A = \int_{\pi/6}^{7\pi/6} (\sqrt{3} \sin x - \cos x) dx$$

$$= \left[-\sqrt{3} \cos x - \sin x \right]_{\pi/6}^{7\pi/6}$$

$$= - \left[\left(\sqrt{3} \cos \frac{7\pi}{6} + \sin \frac{7\pi}{6} \right) - \left(\sqrt{3} \cos \frac{\pi}{6} + \sin \frac{\pi}{6} \right) \right]$$

$$= - \left[\left(\frac{-\sqrt{3} \times \sqrt{3}}{2} + \left(-\frac{1}{2} \right) \right) - \left(\frac{\sqrt{3} \times \sqrt{3}}{2} + \frac{1}{2} \right) \right] = - \left(\frac{3}{2} - \frac{1}{2} \right) - \left(\frac{3}{2} - \frac{1}{2} \right)$$

$$= - \left[-2 - 2 \right] = 4$$

$$= 4 \text{ u}^2$$

Question 8

$$a) i) f(x) = ae^x + 3$$

$$\text{sub } (0, 3.5)$$

$$f(0) = ae^0 + 3 = 3.5$$

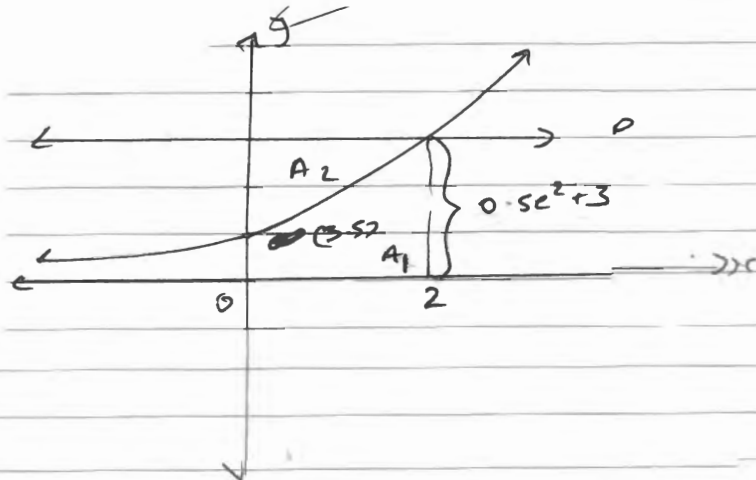
$$\therefore a(1) = 0.5 \quad a \leq 0$$

$$ii) \text{ sub } x=2$$

$$f(2) = 0.5e^2 + 3$$

$$\therefore \text{the y value} = 0.5e^2 + 3$$

iii)



$$A_{\text{Rectangle}} = (e^2 + 6)e^2$$

$$A_2 = A_{\text{Rectangle}} - A_1$$

$$A_1 = \int_0^2 0.5e^x + 3 \, dx$$

$$= \left[0.5e^x + 3x \right]_0^2$$

$$= 0.5e^2 + 6 - 0.5 = 0.5e^2 + 5.5$$

$$A_2 = e^2 + 6 - (0.5e^2 + 5.5)$$

$$= 0.5e^2 + 0.5$$

$$= 0.5(e^2 + 1) \quad u^2$$

$$\therefore \text{shaded region} = 0.5(e^2 + 1) \quad u^2$$

$$Q9) c) K = \$30000$$

$$K = 1 + \frac{9\%}{12} = 1.0075$$

$$n = n$$

$$K = 9250$$

$$i) A_1 = LK$$

$$= 30000(1.0075) = 30225$$

$$ii) A_2 = LK^2 - P$$

$$A_3 = LK^3 - PK - P$$

$$A_1 = LK^4 - P(1 + K + K^2)$$

$$A_1 = LK^4 - \frac{P(K^4 - 1)}{K - 1}$$

$$= LK^4 - \frac{PK^4}{K-1} + \frac{P}{K-1}$$

$$A_1 = K^4(L - \frac{P}{K-1}) + \frac{P}{K-1} = 3333\frac{1}{3} - 3333\frac{1}{3}(1.0075)^4$$

Q 60

iii) from ii)

$$A_n = Lk^n - P \frac{1 - k^n}{1 - k} \quad \text{if } k \neq 1$$

$$\Rightarrow \frac{dA_n}{dn} = \frac{-Lk^n - P}{k-1} = \frac{-P}{k-1}$$

$$k^n = \frac{-P}{Lk - L - P}$$

$$k^n = 10$$

$$n \ln k = \ln(10)$$

$$n = \frac{\ln(10)}{\ln k} = 309$$

$$\therefore n = 309 \text{ months}$$

She will have no money after 309 months

i) $P = P_0 e^{kt}$

$$\text{at } t=0, P=300$$

$$\therefore P_0 = 300$$

$$\text{at } t=100, P=550$$

$$550 = 300 e^{k(100)}$$

$$\ln\left(\frac{550}{300}\right) = k(100)$$

$$k = \frac{\ln\left(\frac{550}{300}\right)}{100} = 0.00606(35803)$$

$$\therefore P_0 = 300$$

$$k = 0.00606(35803)$$

ii) $k + P = 1000$

$$\frac{1000}{300} = \frac{300}{300} e^{kt}$$

$$\frac{10}{3} = e^{kt}$$

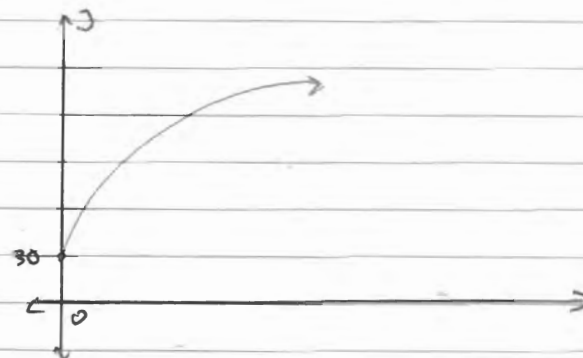
$$\ln\left(\frac{10}{3}\right) = kt$$

$$t = \frac{\ln\left(\frac{10}{3}\right)}{k}$$

$$t = 198.6308$$

$$\therefore t = 199 \text{ days}$$

c)



Q10 a)

$$y = x \ln|x|$$

$$\text{let } u = x \quad u' = 1$$

$$v = \ln|x| \quad v' = \frac{1}{x}$$

$$y' = \ln|x| + 1$$

$$\frac{dy}{dx} = \ln|x| + 1$$

$$\int dy = \int (\ln|x| + 1) dx$$

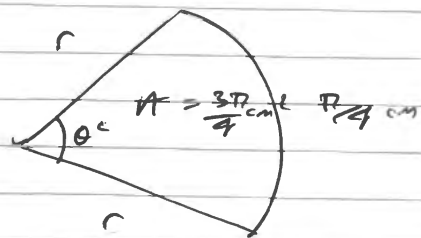
$$y = \int \ln|x| dx + x$$

$$y - x = \int \ln|x| dx$$

$$x \ln|x| - x = \int \ln|x| dx$$

$$\therefore \int \ln|x| dx = x \ln|x| - x + C$$

b)



$$A = \frac{1}{2} r^2 \theta \quad (2) \quad L = r\theta$$

$$\frac{\pi}{4} = r\theta$$

$$\theta = \frac{\pi}{4r} \quad (1)$$

sub (1) into (2)

$$\frac{3\pi}{4} = \frac{1}{2} r^2 \frac{\pi}{4r}$$

$$3\pi = \frac{1}{2} r\pi$$

$$6 = r$$

$$\begin{cases} r = 6 \\ \theta = \frac{\pi}{24} \end{cases}$$

$$c) i) V = 20\pi m^3$$

$$2\pi r^2 \rightarrow 10/m^2$$

$$2\pi r h \rightarrow 8/m^2$$

ii) - - -

$$C = 10(2\pi r^2) + 8(2\pi r h)$$

$$C = 20\pi r^2 + 16\pi r h \quad QED$$

ii) ~~not a min~~

$$V = \pi r^2 h$$

$$20\pi = \pi r^2 h$$

$$h = \frac{20\pi}{\pi r^2}$$

$$\therefore h = \frac{20}{r^2} \quad QED$$

iii) sub h into C

$$C = 20\pi r^2 + 16\pi r \left(\frac{20}{r^2} \right)$$

$$C = 20\pi r^2 + \frac{320\pi}{r}$$

$$C = 20\pi r^2 + 320\pi r^{-1}$$

$$\frac{dC}{dr} = 40\pi r - 320\pi r^{-2}$$

$$\text{Let } \frac{dC}{dr} = 0 \quad (1P)$$

$$0 = 40\pi r - \frac{320\pi}{r^2}$$

$$40\pi$$

$$0 = r - \frac{8}{r^2}$$

$$0 = \frac{r^3 - 8}{r^2}$$

$$r \neq 0$$

$$r = 2$$

$$h = 5$$

1st Deriv test:

+4	2	2.1
\	-	/

$\therefore$ at $r = 2$, it is a minimum

$$\therefore r = 2, h = 5$$

$$\therefore \begin{cases} r = 2m \\ h = 5m \end{cases}$$